教學實作：擴充 DHT11 溫濕度感測器功能

步驟 1：修改硬體定義提示詞 (Devices Definition)

讓 AI 認識 DHT11 感測器及其物理特性。

```
DHT11 Temperature & Humidity Sensor: Pin 20
- Measures: temperature (°C) and relative humidity (%)
- Read mode: single trigger, returns two integer values
- Temperature range: 0–50 °C
- Humidity range: 20–90 % RH
- Physical Rules: Values are integers. Sensor requires ~1 s between reads.
```

步驟 2：定義工具 JSON 規範 (Tools Definition)

在 toolsDefinition 中加入 /dht11 指令的格式規範。

Reading the DHT11 temperature and humidity sensor:

```
{
  "type": "tool_call",
  "method": "/dht11",
  "params": {
    "pin": "<Device pin number. If the user does not specify a pin, ask first.>"
  }
}
```

Success response:

```
{
  "status": "success",
  "method": "dht11",
  "pin": 20,
  "temperature": 26,
  "humidity": 65
}
```

Error response:

```
{
  "status": "error",
  "reason": "dht11_read_failed",
  "pin": 20
}
```

步驟 3：底層 C++ 程式碼實作

在專案中新增 tool_dht11 驅動函數。

```cpp
#include "DHT.h"
#define DHTPIN 20
#define DHTTYPE DHT11
DHT dht(DHTPIN, DHTTYPE);

// Read temperature and humidity from a DHT11 sensor.
// Returns a JSON result string for the agent workflow.
String tool_dht11(int pin) {
  float h = dht.readHumidity();
  float t = dht.readTemperature();

  // Check if any reads failed and exit early (to try again).
  if (isnan(h) || isnan(t)) {
    return "{\"status\":\"error\","
           "\"reason\":\"dht11_read_failed\","
           "\"pin\":" + String(pin) + "}";
  }

  return
    "{\"status\":\"success\","
    "\"method\":\"dht11\","
    "\"pin\":"         + String(pin)  + ","
    "\"temperature\":" + String(t) + ","
    "\"humidity\":"    + String(h) + "}";
}

void setup() {
  // ... 原有初始化內容 ...
  delay(1000);  // 等待感測器穩定
  dht.begin();
}
```

步驟 4：更新工具分發器邏輯 (executeTool)

將 /dht11 方法註冊進系統的分發流程。

```
void executeTool(String command, JsonObject params, bool reCheck = true) {

  if (command == "/digitalwrite" || command == "/analogwrite") {
    // ... 原有內容不變 ...

  }
  else if (command == "/dht11") {
    int pin = params["pin"].as<int>();

    String response = tool_dht11(pin);

    historicalMessages += buildGeminiMessage("user", command, 1);
    historicalMessages += buildGeminiMessage("model", response, 1);
    storeHistoricalMessagesToFile();

    executeToolHistory += command + "," + String(pin) + ", " + response  + ")\n";
    evaluateWorkflowContinuation(reCheck);

  }
  else if (command == "/digitalread" || command == "/analogread") {
    // ... 原有內容不變 ...
  }
  // ... 其餘 tools 不變 ...
}
```